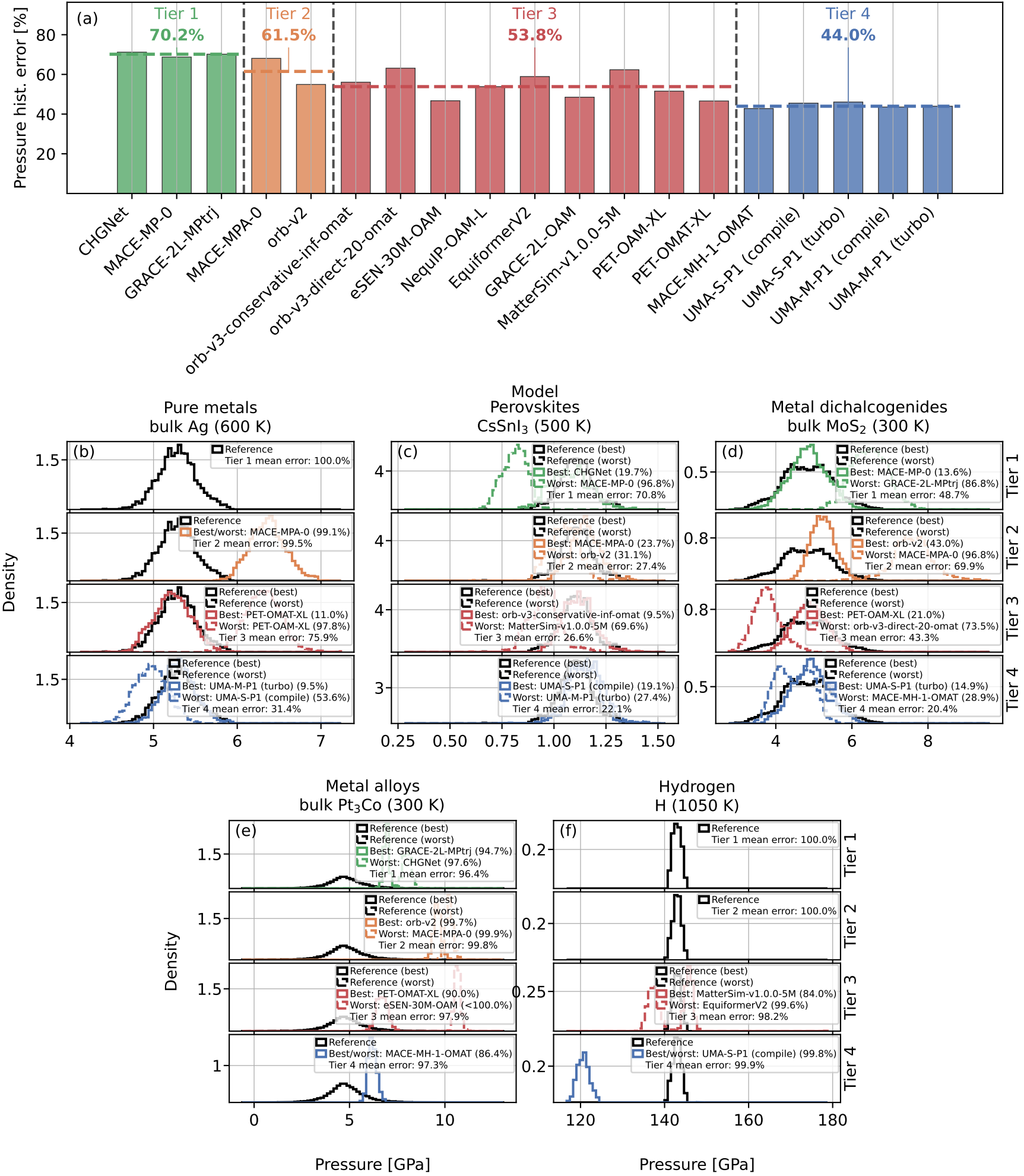

(a)
Tier 1
70.2%
Tier 2
61.5%
Tier 3
53.8%
Tier 4
44.0%
Pressure hist. error [%]
CHGNet
MACE-MP-0
GRACE-2L-MPtrj
MACE-MPA-0
orb-v2
orb-v3-conservative-inf-omat
orb-v3-direct-20-omat
eSEN-30M-OAM
NequIP-OAM-L
EquiformerV2
GRACE-2L-OAM
MatterSim-v1.0.0-5M
PET-OAM-XL
PET-OMAT-XL
MACE-MH-1-OMAT
UMA-S-P1 (compile)
UMA-S-P1 (turbo)
UMA-M-P1 (compile)
UMA-M-P1 (turbo)
Model
Pure metals
bulk Ag (600 K)
Perovskites
CsSnI3 (500 K)
Metal dichalcogenides
bulk MoS2 (300 K)
(b)
Reference
Tier 1 mean error: 100.0%
Reference
Best/worst: MACE-MPA-0 (99.1%)
Tier 2 mean error: 99.5%
Reference (best)
Reference (worst)
Best: PET-OMAT-XL (11.0%)
Worst: PET-OAM-XL (97.8%)
Tier 3 mean error: 75.9%
Reference (best)
Reference (worst)
Best: UMA-M-P1 (turbo) (9.5%)
Worst: UMA-S-P1 (compile) (53.6%)
Tier 4 mean error: 31.4%
(c)
Reference (best)
Reference (worst)
Best: CHGNet (19.7%)
Worst: MACE-MP-0 (96.8%)
Tier 1 mean error: 70.8%
Reference (best)
Reference (worst)
Best: MACE-MPA-0 (23.7%)
Worst: orb-v2 (31.1%)
Tier 2 mean error: 27.4%
Reference (best)
Reference (worst)
Best: orb-v3-conservative-inf-omat (9.5%)
Worst: MatterSim-v1.0.0-5M (69.6%)
Tier 3 mean error: 26.6%
Reference (best)
Reference (worst)
Best: UMA-S-P1 (compile) (19.1%)
Worst: UMA-M-P1 (turbo) (27.4%)
Tier 4 mean error: 22.1%
(d)
Reference (best)
Reference (worst)
Best: MACE-MP-0 (13.6%)
Worst: GRACE-2L-MPtrj (86.8%)
Tier 1 mean error: 48.7%
Reference (best)
Reference (worst)
Best: orb-v2 (43.0%)
Worst: MACE-MPA-0 (96.8%)
Tier 2 mean error: 69.9%
Reference (best)
Reference (worst)
Best: PET-OAM-XL (21.0%)
Worst: orb-v3-direct-20-omat (73.5%)
Tier 3 mean error: 43.3%
Reference (best)
Reference (worst)
Best: UMA-S-P1 (turbo) (14.9%)
Worst: MACE-MH-1-OMAT (28.9%)
Tier 4 mean error: 20.4%
Tier 1
Tier 2
Tier 3
Tier 4
Density
Metal alloys
bulk Pt3Co (300 K)
Hydrogen
H (1050 K)
(e)
Reference (best)
Reference (worst)
Best: GRACE-2L-MPtrj (94.7%)
Worst: CHGNet (97.6%)
Tier 1 mean error: 96.4%
Reference (best)
Reference (worst)
Best: orb-v2 (99.7%)
Worst: MACE-MPA-0 (99.9%)
Tier 2 mean error: 99.8%
Reference (best)
Reference (worst)
Best: PET-OMAT-XL (90.0%)
Worst: eSEN-30M-OAM (<100.0%)
Tier 3 mean error: 97.9%
Reference
Best/worst: MACE-MH-1-OMAT (86.4%)
Tier 4 mean error: 97.3%
(f)
Reference
Tier 1 mean error: 100.0%
Reference
Tier 2 mean error: 100.0%
Reference (best)
Reference (worst)
Best: MatterSim-v1.0.0-5M (84.0%)
Worst: EquiformerV2 (99.6%)
Tier 3 mean error: 98.2%
Reference
Best/worst: UMA-S-P1 (compile) (99.8%)
Tier 4 mean error: 99.9%
Pressure [GPa]
Pressure [GPa]